GEOPHYSICAL RESEARCH LETTERS, VOL. 8, NO. 8, PAGES 880-882, AUGUST 1981

PALEOMAGNETISM OF THE LATE ORDOVICIAN NEDA IRON ORE FROM WISCONSIN, IOWA AND ILLINOIS

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Abstract. The Neda Iron Formation (latest Ordovician) is a hematite and goethite rich oolitic ore which occurs in lens shaped deposits on top of the Maquoketa shale (Late Ordovician), at only a few locations in Wisconsin, Iowa and Illinois. Thermal demagnetization of samples from these three states, indicates the remanence is carried by hematite. Chemical demagnetization suggests the remanence is produced by the interstitial material rather than the oolites.

The paleopole averaged for all sites from 25 samples of the Neda ore is at S45.4°, W48° ($\alpha 95$ = 16°). This pole position is similar to late Mississippian to early Permian of North America rather than latest Ordovician. The difference between the known age of the Neda and the time of magnetization suggests that the hematite in the ore was produced from dehydration of goethite probably as a result of tectonic uplift in Wisconsin during late Mississippian time. In addition, these results add evidence to the theory that the Neda ore was formed in two stages. The oolites were produced in a near shore environment during the shoaling of the Ordovician seas. The oolites were then incorporated in iron rich muds. Burial and later uplift dehydrated the iron hydroxide to hematite.

Introduction

A major paleomagnetic study of paleozoic rocks from the Wisconsin area is underway at The University of Wisconsin-Milwaukee. The stratigraphic section of initial interest is the Ordovician-Silurian boundary, which includes the Maquoketa Shale (Late Ordovician), the Neda Iron Formation (latest Ordovician) and the Mayville Dolomite (Early Silurian). The paleomagnetic results from studies of the Neda Iron Formation will be presented in this report.

Geology

The Neda Iron ore is a hematite and goethite rich oolitic formation exposed in only a few locations in eastern Wisconsin, northeastern Iowa and northern Illinois. The locations of the major outcrops are given in Figure 1. The most extensive exposures of the Neda are at Neda, Wisconsin, where the ore had been mined until the early 1900's. It generally occurs in lens shaped deposits on top of the Maquoketa Shale and is usually about one meter thick. However, at Neda, Wisconsin, a 13.4 meter section has been measured in the underground workings (Rosenzweig, 1951).

Originally the Neda was considered to be of Clinton age because of its similar appearance to the Silurian Clinton ores of New York. Today it is considered latest Ordovician based on the Late Ordovician fossils found in it (Savage and Ross, 1916), and the unconformable relation of the Neda with the Silurian Mayville Dolomite which overlies it.

The mineral content of the Neda is primarily iron oxides with goethite the most common mineral in the oolites and hematite the main component of the matrix (Hawley and Beavan, 1934).

A major uncertainty concerning the Neda is the origin of the iron and its time of enplacement. All evidence indicates that the iron hydroxides were present before the transgression of the Silurian seas. Paull (1977) suggests that the ooliths were replaced with iron oxide before they were incorporated in the Maquoketa mud, because hematitic ooliths have been found between bedding planes of the Maquoketa shale at the base of the Neda. Hawley and Beavan (1934) prefer an origin which involves growth of most of the ooliths from colloidal ferric hydroxide in a quiet environment, and at the same time, others were washed in from similar deposits formed nearby.

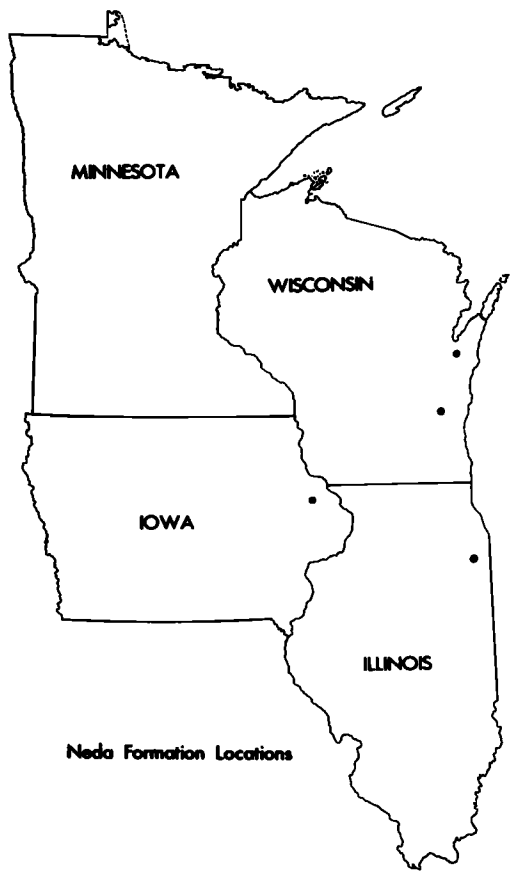


Fig. 1. Location map for Neda ore samples. Neda, Wi. (44.16°N, 88.08°W); Dubuque, Iowa (42.5°N, 90.75°W); Kankakee, Ill. (41.1°N, 87.8°W).

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Paper number 1L1022.
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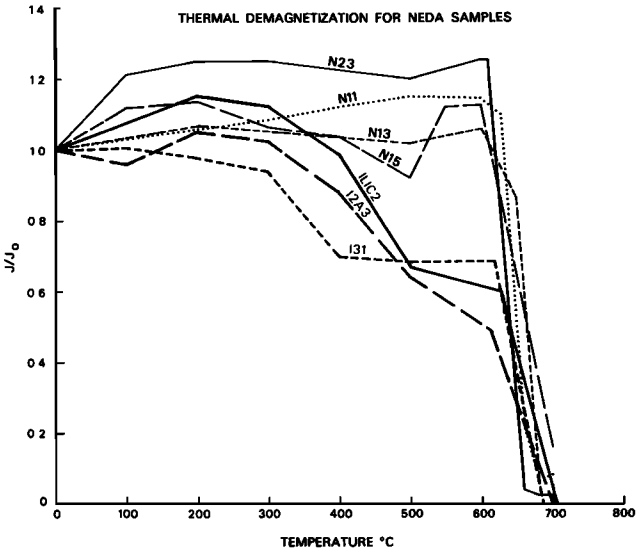


Fig. 2. Zijdervelt plot of chemical demagnetization for sample ILIC from Kankakee, Illinois.

Magnetic Methods

A total of 25 hand samples were collected at three of the Neda outcrops; 12 at Neda, Wisconsin; 7 at Dubuque, Iowa; and 6 at Kankakee, Illinois. A minimum of two samples and a maximum of six samples were cut from each hand sample for paleomagnetic studies.

Magnetic measurements were made using a two axes cryogenic magnetometer. Four samples from each site were subjected to A.F. demagnetization to 100 mT and chemical demagnetization for 300 hrs. in 6 N HCl. The remaining samples were thermally demagnetized stepwise to 700° C in air.

Results

A.F. demagnetization to 100 mT had no major effect on any of the samples. Most samples showed a 5-10% decrease in intensity but no change in direction.

Chemical demagnetization up to 300 hours in 6 N HCl did produce decreases in intensity for some of the samples. However, none of the samples exhibited any major changes in direction. The samples from Illinois gave the greatest decreases in intensity of up to 60%; samples from Iowa gave decreases of about 20%; whereas those from Neda, Wisconsin changed by only 5%. A typical Zijdervelt plot for a sample from Illinois is given in Figure 2.

The differences in chemical demagnetization at the three sites reflect differences in the matrix materials. The samples from Wisconsin have the least amount of dolomite in the matrix whereas those from Illinois have the highest amount. Only the samples from Illinois showed any noticeable color change with chemical demagnetization. They changed from orange-red to white. Chemical leaching was terminated after 300 hours because the samples became too friable to handle. The matrix became very soft, and for some samples only a residue of red colored ooliths remained.

Thermal demagnetization provided the most information as to the carrier of the remanent magnetization. For all samples remanence was removed by 700° C. The samples from Wisconsin consistently showed a rise in intensity at 100-150° C and a reversal of the magnetic direction from normal to reversed. From 150° C to 650° C, the remance direction remained constant; above 650° C the remanence was destroyed as indicated in Figure 3. Vector subtraction of the NRM from the direction obtained after 150° C indicated a difference vector equivalent to the present day field. The final remanent direction is given in Table 1. The samples from Iowa and Illinois gave consistently reverse magnetic directions. However, in most samples from these two sites the intensity started to decrease between 400 and 500° C but there was no significant change in magnetic direction up to 650° C, after which the remanence was destroyed (Figure 3). The remanent directions from these sites are also given in Table 1. The average declination and inclination for all samples is (152°, -12°), which gives a V.G.P. at (S45.4°, W48°).

Discussion

The limited results from the chemical demagnetization studies would suggest that the remanence is carried primarily by the iron oxide incorporated in the matrix of the ore. The samples from Illinois lost most of their red color with leaching and Zijderweld plots distinctly showed a linear decrease toward the origin, before they disintegrated. In addition, many of the oolites that remained still maintained their red color.

Thermal demagnetization indicates that the main carrier of the remanence is hematite, since in all samples the remanence remained fairly constant until the Curie temperature of hematite (680°). The present day overprint which the Wisconsin samples exhibit is probably the result of surface weathering and the reconversion to goethite, because the overprint is lost at 120° C. The drop in intensities at 400-500° C for samples from Iowa and Illinois suggest that the matrix material is probably finer grained.

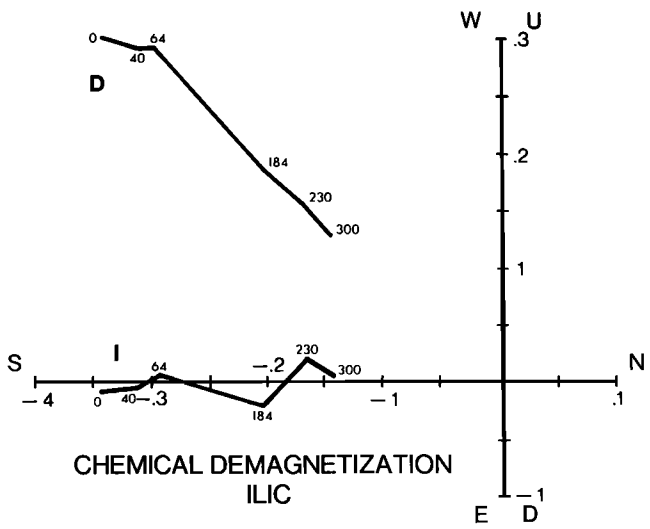


Fig. 3. Thermal demagnetization curves for Neda samples from Wisconsin, Iowa and Illinois.

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Table 1. Paleopole positions for the three Neda Ore sites

Location	N	Dec.	Inc.	α(95)	k	V.G.P. Lat.	Long.
Neda, Wisconsin	12	162	-6.7	5.3	67	S46.8	W52
Dubuque, Iowa	7	146	-18	5.6	120	S46	W38
Kankakee, Illinois	6	148	-11	20	45	S43	W34
Average of three sites		152	-12	16°	200	S45.4	W48

There appears to be only one dominant paleomagnetic direction discernable in the Neda ores from the three sites studied. This direction (S45.4°, W48°) is not consistent with published poles for the Ordovician or Silurian of North America, but rather it is a Late Paleozoic pole more consistent with Permian time. It is almost identical with the direction of secondary magnetization which Van der Voo and French (1977) found in Ordovician sediments from the folded Appalachians.

These results would suggest two possibilities for the origin of the matrix of the ore. Either hematite was precipitated directly from iron rich waters, and later recrystallization produced grains large enough to maintain a remanence, or the matrix was originally goethite which later dehydrated to hematite. In either case, the remanence is interpreted as a chemical remanence associated with an uplift in Wisconsin during late Mississippian to Permian time.

Recent x-ray analysis by Synowiec (1980) and Brown (1981, personal communication) has indicated the dominant iron mineral of the whole rock is goethite with lesser amounts of hematite.

These results may provide new information relating to the origin of the Neda Iron ore. Because the oolites do not leach as readily as the matrix and do not seem to influence the remanence, one could conclude that they were formed earlier than the matrix. Either contemporaneously or shortly after, the oolites were incorporated in iron rich muds. The ultimate source of the iron rich muds could have been either from the Queenston delta from the east, as suggested by Templeton and Willman (1963), or from the weathering of crystalline rocks to the northwest. The iron was originally in the form of goethite which later converted to hematite. Preliminary paleomagnetic studies of the Queenston Shale by Kean (1980), indicate similar pole positions for the Queenston and the Neda. However, this alone does not conclusively argue for an eastern source, because the Queenston Shale could have undergone the same uplift and dehydration as areas to the west.

Conclusions

Paleomagnetic studies on Neda Iron ore from sites in Wisconsin, Iowa and Illinois give a paleopole position at (S45.4°, W48°) which is considered Permian in age rather than Ordovician. Chemical demagnetization studies suggest the remanence is carried in the matrix of the ore rather than by the oolites; thermal demagnetization indicates the remanence is produced by hematite. The difference between the known age of the Neda and the time of magnetization suggests that the hematite in the ore was produced from dehydration of goethite probably as a result of the tectonic uplift in Wisconsin during late Mississippian to early Permian time.

These results add credence to the theory that the Neda ore was formed in two stages. The oolites were produced in a near shore environment during the shoaling of the Ordovician seas. The oolites were then incorporated in iron rich muds either from the eastern Queenston delta or from the western trans-continental arch. Burial and later uplift, dehydrated iron hydroxide to hematite.

Acknowledgments. The author would like to thank Drs. Bruce Brown, Emily Giffin and Richard Paull for their helpful discussions. This project was supported by N.S.F. Grant EAR79-20073.

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(Received May 29, 1981;
accepted June 16, 1981.)

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